

Brain Tumour Detection Using EfficientNet

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April 27, 2026

Abstract

Brain tumours are critical neurological disorders that require timely and accurate diagnosis for effective treatment. Manual MRI analysis is often time-consuming and subject to human variability. In this project, we develop a deep learning-based classification system using EfficientNet architectures to automatically detect brain tumours. The model classifies MRI images into four categories: glioma, meningioma, pituitary tumour, and no tumour. We incorporate preprocessing techniques, class balancing, and transfer learning to improve performance. Experimental results demonstrate that EfficientNet-based models outperform traditional CNN approaches in accuracy and robustness. This system aims to assist medical professionals by providing a reliable and efficient diagnostic tool.

1 Introduction

Brain tumours significantly impact human health and can lead to severe complications if not detected early. Accurate classification of tumour types is essential for treatment planning and improving survival rates. However, manual analysis of MRI scans depends heavily on radiologist expertise and can be prone to inconsistencies.

Recent advancements in deep learning, particularly Convolutional Neural Networks (CNNs), have revolutionized medical image analysis. Among these, EfficientNet has shown superior performance due to its compound scaling approach, which balances depth, width, and resolution efficiently.

In this project, we propose a deep learning-based framework using CNN and EfficientNet architectures to classify brain tumour MRI images into four categories. The system leverages preprocessing techniques and transfer learning to achieve high accuracy and generalization.

2 Literature Review

Traditional approaches for brain tumour classification relied on handcrafted feature extraction and classical machine learning algorithms. These methods often lacked robustness and failed to generalize across datasets.

Recent works have explored deep CNN architectures for automated tumour detection. Models like VGG, ResNet, and DenseNet have demonstrated promising results but often require large computational resources.

EfficientNet introduced a novel compound scaling method, improving accuracy while maintaining efficiency. Several studies have applied EfficientNet for medical imaging tasks with strong performance.

In this project, we extend these approaches by:

- Using grayscale MRI preprocessing to reduce computational complexity
- Applying class-weighted loss to handle balanced but sensitive classification
- Implementing multi-stage training (freeze + fine-tuning)
- Comparing multiple architectures (CNN, EfficientNet-B3, EfficientNet-V2-S)

Unlike many existing works that rely on a single model, our approach provides a comparative study across architectures, highlighting performance improvements using transfer learning.

3 Proposed Methodology

3.1 Data Preprocessing

1. All images are converted from RGB to grayscale to reduce dimensionality and focus on structural features.
2. Training, validation, and test transforms are applied for normalization and augmentation.

3.2 Dataset Split

- Training Size: 4480
- Validation Size: 1120
- Test Size: 1600

3.3 Class Distribution

- Classes: glioma, meningioma, no_tumor, pituitary
- Each class contains 1400 images

3.4 Image Resizing

All images are resized to a fixed dimension for model compatibility.

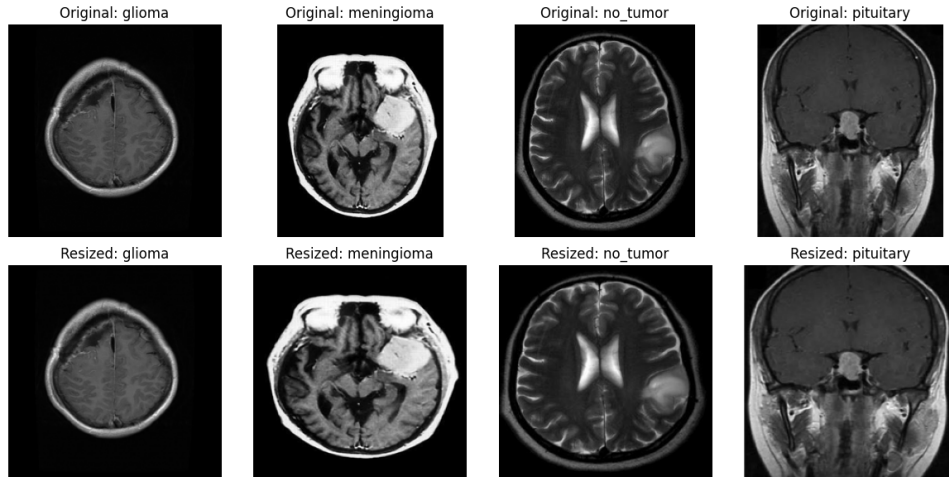


Figure 1: Original vs Resized Images

3.5 Model 1: CNN

- Loss Function: Binary Cross Entropy (BCE)
- Optimizer: Adam
- Training: 50 epochs

3.6 Model 2: EfficientNet-B3

- Backbone frozen (only classifier trained)
- Loss: BCE with class weights
- Optimizer: Adam
- Training:
 - Frozen Backbone and training (Feature Extraction): 10 epochs
 - Fine-tuning: Unfreeze last layers and retrain
- Testing

3.7 Model 3: EfficientNet-V2-S

- Backbone frozen initially
- Loss: BCE with class weights
- Optimizer: Adam
- Training:
 - Frozen Backbone and training (Feature Extraction): 10 epochs
 - Fine-tuning with unfrozen layers
- Testing

4 Experimental Results

4.1 Dataset

The dataset used is the **Brain Tumor MRI Dataset** obtained from Kaggle.

4.2 CNN Model Results

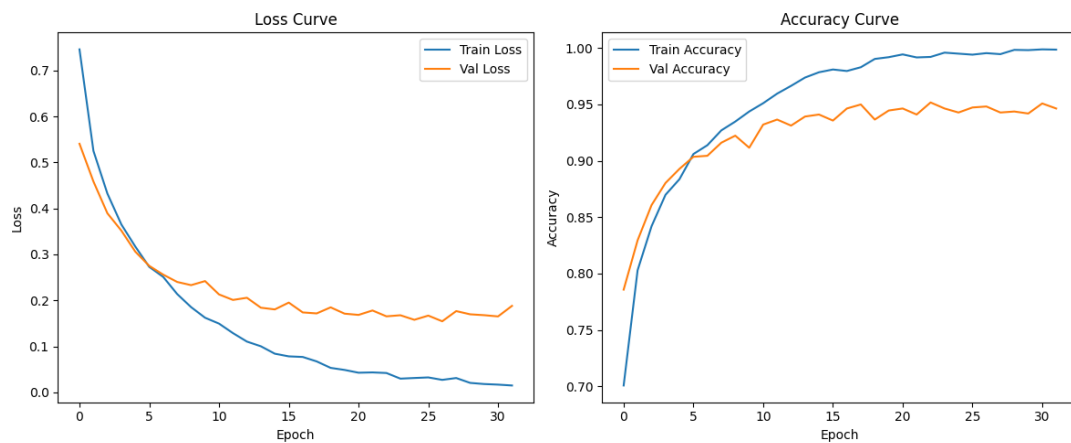


Figure 2: CNN Loss and Accuracy Curve

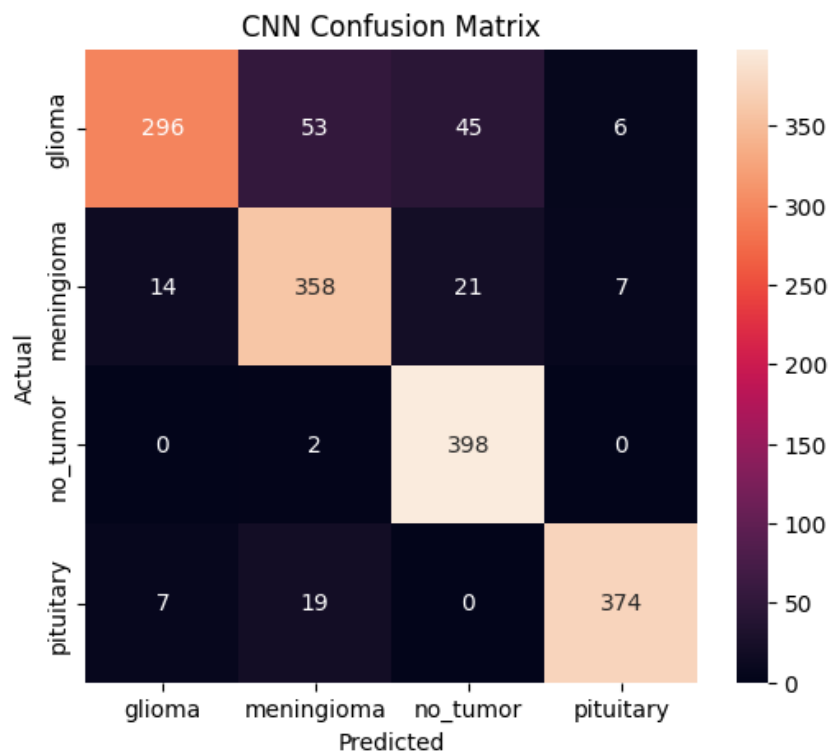


Figure 3: CNN Confusion Matrix

4.2.1 CNN Classification Report

Accuracy: 0.8912

Classification Report:

	precision	recall	f1-score	support
glioma	0.93	0.74	0.83	400
meningioma	0.83	0.90	0.86	400
no_tumor	0.86	0.99	0.92	400
pituitary	0.97	0.94	0.95	400
accuracy			0.89	1600
macro avg	0.90	0.89	0.89	1600
weighted avg	0.90	0.89	0.89	1600

4.3 EfficientNet-B3 Results

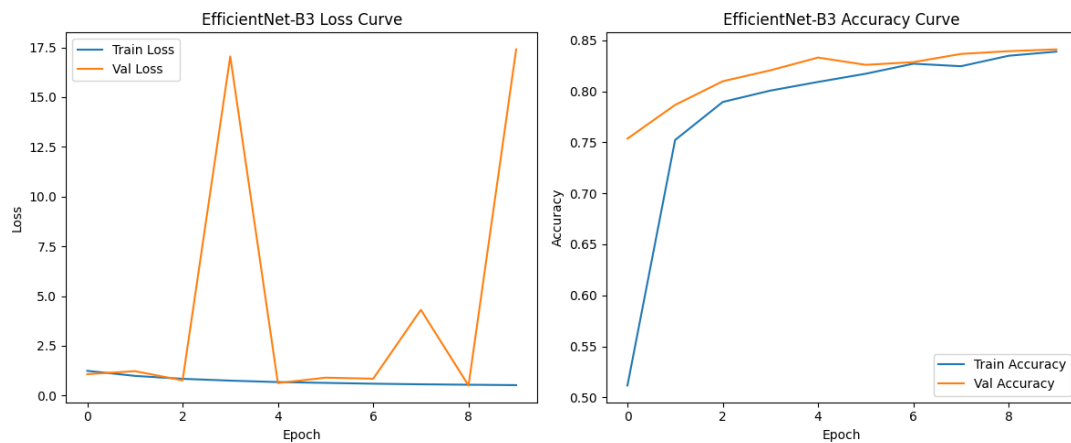


Figure 4: EfficientNet-B3 Loss and Accuracy Curve

4.3.1 EfficientNet-B3 Classification Report

Accuracy: 0.9387

Classification Report:

	precision	recall	f1-score	support
glioma	0.98	0.81	0.88	400
meningioma	0.89	0.97	0.93	400
no_tumor	0.92	1.00	0.95	400
pituitary	0.98	0.98	0.98	400
accuracy			0.94	1600
macro avg	0.94	0.94	0.94	1600
weighted avg	0.94	0.94	0.94	1600

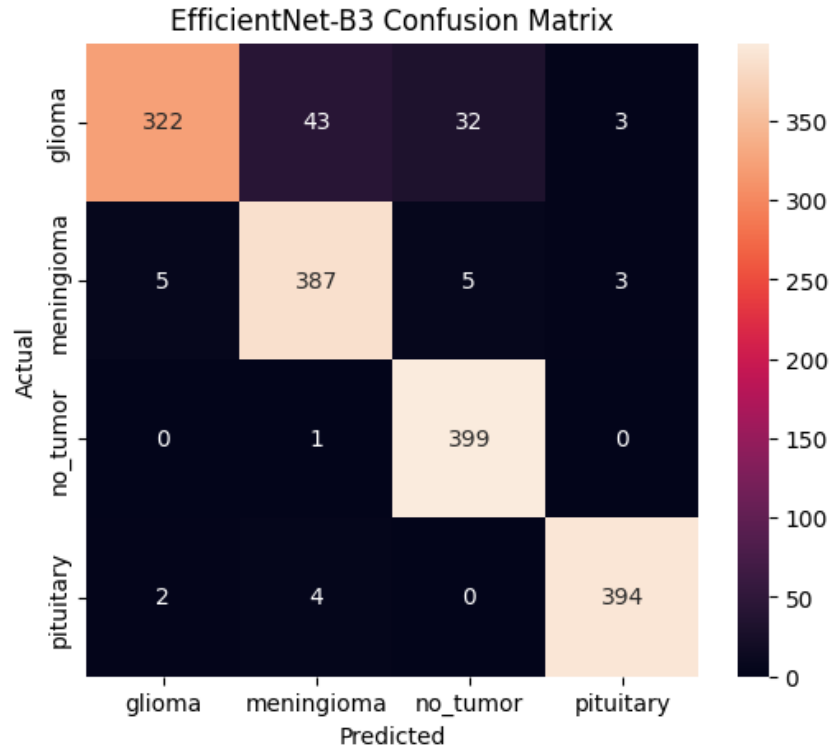


Figure 5: EfficientNet-B3 Confusion Matrix

4.4 EfficientNet V2-S Results

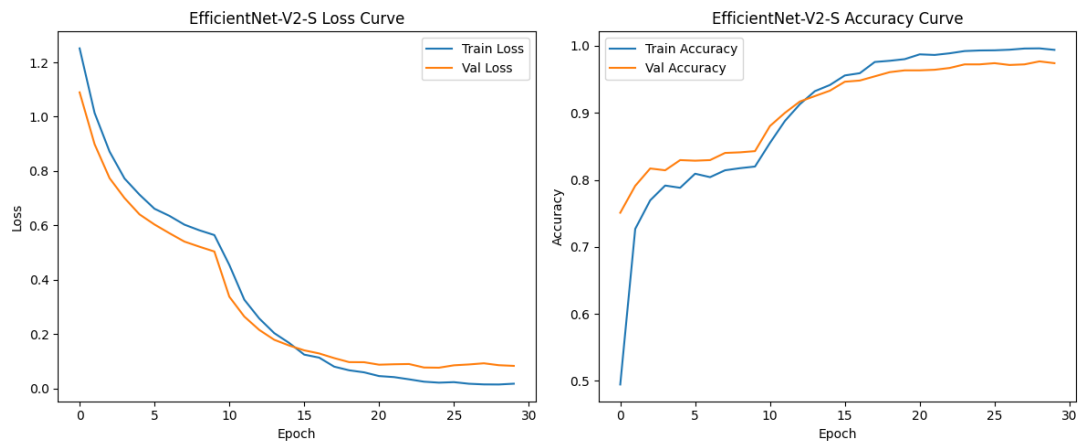


Figure 6: EfficientNet V2-S Loss and Accuracy Curve

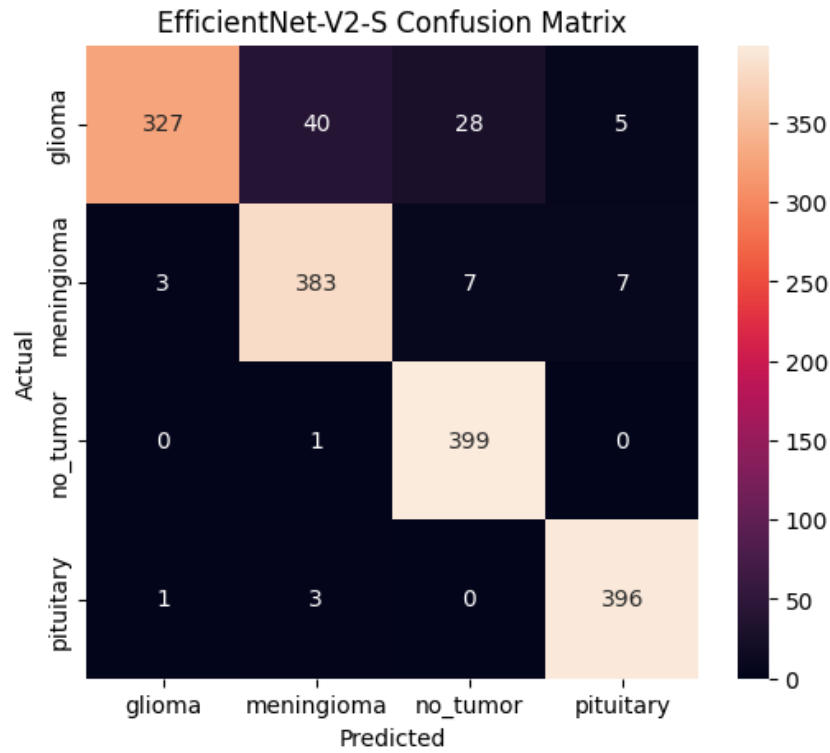


Figure 7: EfficientNet V2-S Confusion Matrix

4.4.1 EfficientNet V2-S Classification Report

Accuracy: 0.9406

Classification Report:

	precision	recall	f1-score	support
glioma	0.99	0.82	0.89	400
meningioma	0.90	0.96	0.93	400
no_tumor	0.92	1.00	0.96	400
pituitary	0.97	0.99	0.98	400
accuracy			0.94	1600
macro avg	0.94	0.94	0.94	1600
weighted avg	0.94	0.94	0.94	1600

4.5 Model Comparison

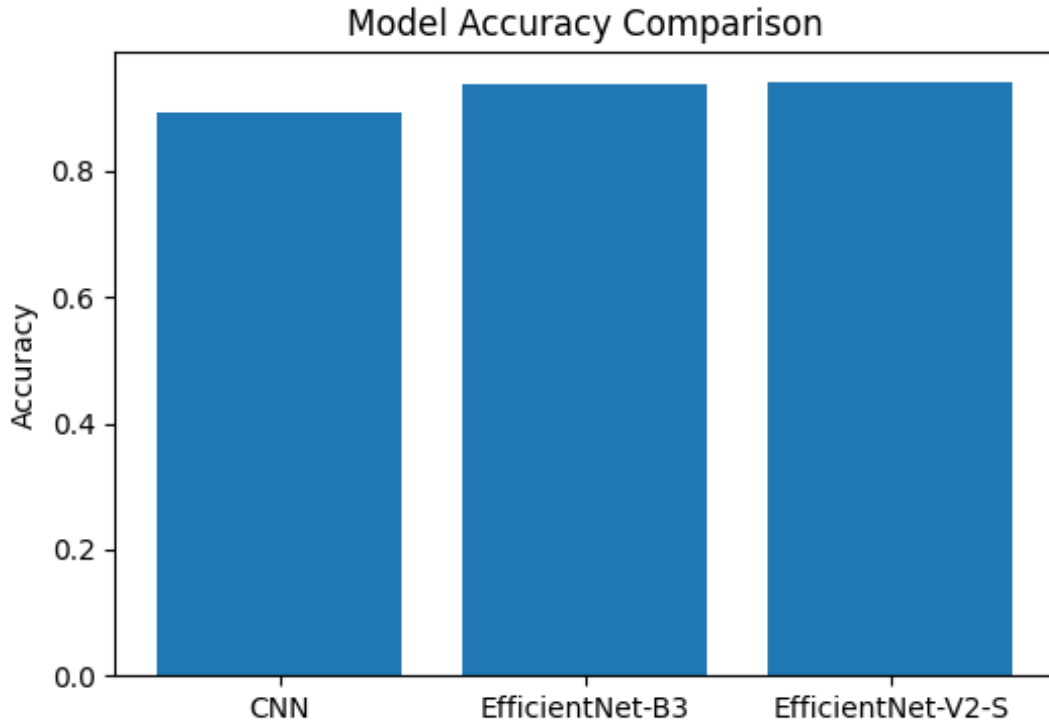


Figure 8: Model Accuracy Comparison

5 Summary

In this project, we developed a deep learning-based system for brain tumour classification using MRI images. We implemented and compared three models: a basic CNN, EfficientNet-B3, and EfficientNet-V2-S.

The results show that transfer learning-based EfficientNet models significantly outperform the traditional CNN approach in both accuracy and generalization. Fine-tuning further improves performance by adapting pretrained features to the medical imaging domain.

Overall, the proposed system demonstrates strong potential for assisting medical professionals in automated tumour detection. Future work can focus on real-time deployment, segmentation tasks, and clinical validation.

6 References

References

- [1] Brain Tumor MRI Dataset, Kaggle. Available at: <https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>
- [2] Mingxing Tan and Quoc V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks", Proceedings of the International Conference on Machine Learning (ICML), 2019.
- [3] Sajjad, M., Khan, S., Muhammad, K., Wu, W., Ullah, A., and Baik, S. W., "Multi-grade brain tumor classification using deep CNN with extensive data augmentation", *Journal of Computational Science*, 30, pp. 174–182, 2019.